

Rayyan Moosani

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

Expected 2029

EXPERIENCE

Software Engineer

May 2026 - Present

XORBIX

- Cut validation time **50x** to **442K rows/sec** via chunked vectorized **pandas** boolean-mask engine
- Slashed **Flask REST API** latency **5.5x** (3.1s to 0.56s) by collapsing 301 **N+1** queries into 3 bulk reads
- Reduced AI extraction time by **95%** (6 min to 5s) in a **C/.NET WPF** app via real-time data streaming
- Generated **200+** files of **.NET/React** apps, cutting setup effort **85%**, via self-made 6-step codegen application

Software Automation Developer

Aug 2024 - Sep 2024

SCITRA

- Built real-time reporting system using **Python Telegram** and **Google Sheets APIs**, reducing latency by **40%** across **30+ lines**
- Engineered **Unity** and **C#** digital twins with **IoT** data, improving predictive maintenance accuracy by **30%**
- Automated fault logging using **Python** pipelines, accelerating engineering resolution time by **25%**

Software Developer

Sep 2025 – Dec 2025

HICUW

- Optimized frontend performance using ES6+ refactoring, reducing bundle size by **20%** and improving load time
- Developed **Python Discord** bots, eliminating manual workflows and increasing efficiency by **35%**
- Built real-time **React** features using event-driven architecture, increasing engagement for **100+ users**

IT Intern

Jun 2024 - Jul 2024

AGMC

- Digitized workflows using **MS Forms** and **SharePoint**, reducing manual processing effort by **30%**
- Migrated legacy systems to **M365**, improving data accessibility and retrieval speed by **40%**
- Resolved **50+ IT tickets** using enterprise tools, improving system reliability and user productivity

PROJECTS

StoryVerse | TypeScript, Node.js, Express, React, PostgreSQL, Drizzle ORM, FFmpeg, OpenAPI, Zod

- Shipped a **portal** where viewers watch AI trailers made by authors and **vote to greenlight** them for **funding**.
- Built **AI pipeline** turning scripts into trailers and shorts across **7+ tiers** of **20+ models**.
- Cut generation cost **85%** by building an **editable, human-in-the-loop** pipeline state machine.
- Raised **character consistency** $\sim 2/9$ to $9/9$ via **prompt-injection** layer locking descriptions.
- Synced multi-scene **narration** and **lip-sync** to video via **FFmpeg** with **R2-backed** storage.

Anr Awaaz | Python, FastAPI, Whisper, PHOIBLE

- Built phoneme analysis pipeline using **Whisper** and **PHOIBLE**, improving feedback accuracy by **30%**
- Designed **LLM** backend using **FastAPI** and Claude APIs, delivering real-time personalized corrections
- Integrated multilingual voice workflows via **REST APIs**, enabling scalable real-time learning

Games.Random | Node.js, Express, MongoDB, Claude API, Docker

- Built a gaming/learning platform using **Node.js** to generate playable 2D games from text with editable code.
- Streamed live code generation using **Server-Sent Events** to achieve real-time rendering.
- Engineered secure auth & deployment using **MongoDB**, **OAuth 2.0**, and **Docker** containers.

TECHNICAL SKILLS

Languages: Python, C#, TypeScript/JavaScript, SQL, C, Racket, XAML, HTML5, CSS3.

AI/ML: LLMs, RAG pipelines, Anthropic Claude API, OpenAI API, Ollama, LM Studio, Claude Code, prompt engineering, Whisper, Pandas, NumPy, PHOIBLE, Selenium.

Frameworks, Infrastructure & APIs: .NET 8, ASP.NET Core, WPF (MVVM), EF Core, FastAPI, Flask, React, Vite, Node.js, Express, TailwindCSS, Docker, PostgreSQL (pgvector), SQLite, Supabase, MongoDB, REST APIs, WebSockets, OAuth 2.0, xUnit, Azure DevOps, CI/CD (Jenkins), Git/GitHub.